

COURSE TITLE : METROLOGY LAB
COURSE CODE : 3308
COURSE CATEGORY : A
PERIODS / WEEK : 4
PERIODS / SEMESTER : 60
CREDIT : 3

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Vernier Calipers , Micrometers	14
2	Gauges , Height Gauges and Angle measuring instruments	14
3	Advanced Measuring Instruments	16
4	Advanced Calibration Equipments	16
	TOTAL	60

COURSE OUTCOME:

Understand and familiarize the use of basic measuring instruments:-

1. Familiar with the use of different types of Vernier Calipers
2. Familiar with the use of different types of Micrometers
3. Familiar with the use of different types of Gauges
4. Familiar with the use of Height Gauges and Angle measuring instruments
5. Basic Measurements of Vernier Caliper And Vernier Height Gauge
6. Basic Measurement of Micrometer
7. Measurement of Bores Using Dial Bore Indicator

Familiarize advanced measuring instruments

8. Electronic Height gauge
9. Profile projector
10. CMM
11. Surface finish tester
12. Toolmakers microscope

Familiarize advanced Calibration Equipments

13. Caliper checker
14. Grade 0 slip gauges
15. Dial calibration tester
16. Optical flat & Optical parallel
17. Monochromatic light unit

(At least 12 Experiments are to be done)

Sl.No	Experiments
1.	Basic Measurements Of Vernier Caliper
2.	Basic Measurement Of Micrometer
3.	Measurement Of Bores Using Dial Bore Indicator
4.	Basic Measurement On Vernier Height Gauge And Digital Height Gauges
5.	Basic Measurement of Gauges
6.	Basic Measurement of Vernier Bevel Protractor
7.	Basic Measurement of Angle Measuring Instruments- Sine Bar, Sine Table And Sine Centre
8.	Basic Measurement of Electronic Height Gauge
9.	Surface Roughness Measurement By Taly Surf

10.	Basic Measurement of Profile Projector
11.	Basic Measurement on Hardness Tester
12.	Basic Measurement of Toolmakers Microscope
13.	Basic Measurement of CMM
14.	Calibration Testing Equipments

1.1.0 Familiar with the use of different types of Vernier Calipers

- 1.1.1 Dial Caliper
- 1.1.2 Digital Caliper
- 1.1.3 Depth Caliper

1.2.0 Familiar with the use of different types of Micrometers

- 1.2.1 Depth Micrometers
- 1.2.2 Flange Micrometers
- 1.2.3 Blade Micrometers
- 1.2.4 Stick Micrometers
- 1.2.5 Ball end Micrometers
- 1.2.6 Three leg Micrometers
- 1.2.7 Point Micrometers

MODULE II

2.1.0 Familiar with the use of different types of Gauges

- 2.1.1 Understand gauges-Slip Gauges, Plug Gauges, Ring Gauges, Feeler Gauges, Thread pitch Gauges, Radius and fillet Gauges, Bore dial Gauges, Pin Gauges, Dial tester indicator

2.2.0 Familiar with the use of Height Gauges and Angle measuring instruments

- 2.2.1 Vernier Height Gauges
- 2.2.2 Digital Height Gauges
- 2.2.3 Vernier bevel protractor
- 2.2.4 Sine bar/Sine table/Sine centre

MODULE III

3.1.0 Familiarize advanced measuring instruments

- 3.1.1 Describe the working principle Electronic Height gauge
- 3.1.2 Describe the Profile projector
- 3.1.3 Describe the working of CMM
- 3.1.4 Understand Surface finish tester
- 3.1.5 Describe the working of Toolmakers microscope

MODULE IV

4.1.0 Familiarize advanced Calibration Equipments

- 4.1.1 Describe Caliper checker
- 4.1.2 Explain Grade 0 slip gauges
- 4.1.3 Describe Dial calibration tester
- 4.1.4 Describe Optical flat & Optical parallel
- 4.1.5 Describe Monochromatic light unit

REFERENCES

- 1 Engineering Metrology by R.K.Jain
- 2 A Text Book of Metrology by M. Mahajan
- 3 Tool and Die Maker by Mahajan
- 4 Engineering Metrology by K. J. Hume
- 5 Engineering Metrology and Instrumentation by R. K. Rajput
- 6 Metrology and Quality Control by P. G. Deshpande